Supporting Information 2

ADDITIONAL RESULTS ON OTHER REACTION EQUILIBRIA

Note on Updates

Future updates or revisions to this document will be made available at: https://github.com/daferro/Notebooks-Chemical_Equilibrium

Table of contents

1. **What is this document about?** 2
2. **The $N_2 + 3H_2 \leftrightharpoons 2NH_3$ equilibrium** 3
3. **The $O_2 + 2H_2 \leftrightharpoons 2H_2O$ equilibrium** 6
4. **References** 9

1. What is this document about?

This document summarizes representative results obtained with ***Notebook 2*** when applied to elementary gas-phase reactions beyond the $\mathbf{N_2O_4 \rightleftharpoons 2NO_2}$ case study. The purpose is twofold: (i) to illustrate the generality of the notebook workflow and (ii) to provide reference outputs that can be used for comparison and teaching. In particular, we analyze the following equilibrium:

- $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$
- $O_2 + 2\,H_2 \rightleftharpoons 2\,H_2O$

2. The $N_2 + 3H_2 \leftrightharpoons 2NH_3$ equilibrium

When running ***Notebook 2***, this reaction is defined by the following input string:

- `N2 + 3 H2 -> 2 NH3`

The remaining input parameters required by the notebook are listed in Table S1.

Table S1. Input data for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. $\Delta_f H^\circ$ is reported in kJ·mol^{-1}, and S_m° and $C_{p,m}^\circ$ in J·mol^{-1}·K^{-1}. These three values correspond to 298 K.[1]

	$\Delta_f H^\circ$	S_m°	$C_{p,m}^\circ$	PubChem CID	Molecular charge	Number of unpaired electrons	Symmetry number
N_2	0.00	191.61	29.125	947	0	0	2
H_2	0.00	130.684	28.824	783	0	0	2
NH_3	-46.11	192.45	35.06	222	0	0	3

For initial amounts equal to the stoichiometric coefficients of the reactants (*i.e.* 1 mol of N_2 and 3 of H_2, and 0 mol of NH_3), the variation of $G(\xi)$ and of $A(\xi)$ with the extent of reaction, ξ, is shown in Figs. S1 and S2, respectively.

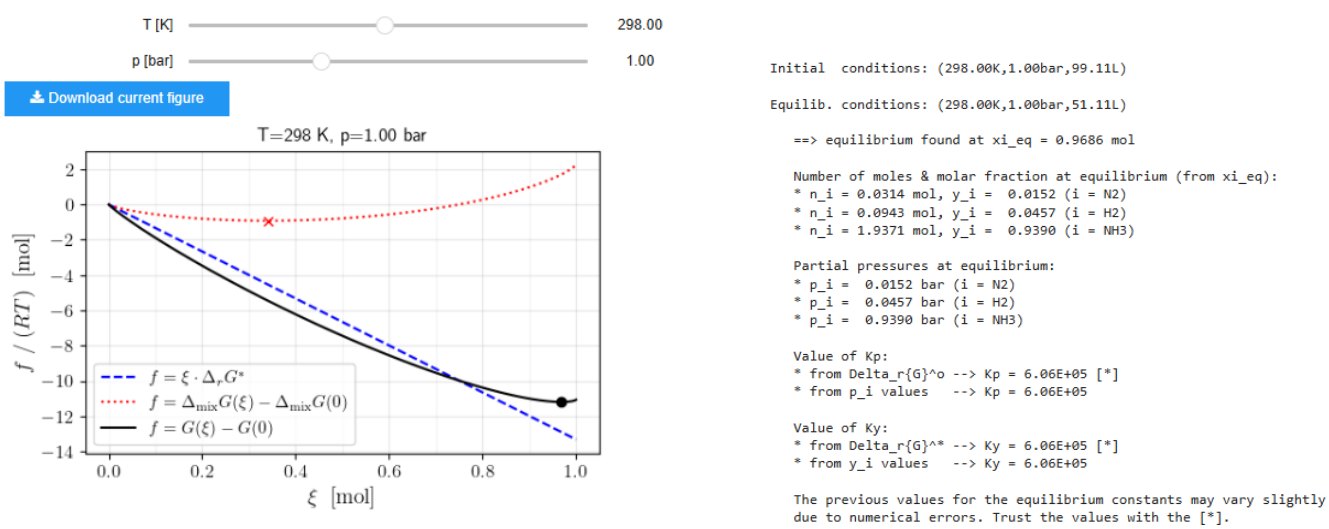


Figure S1. Screenshots showing equilibrium results at constant p and T for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. Moving the sliders updates the plot interactively.

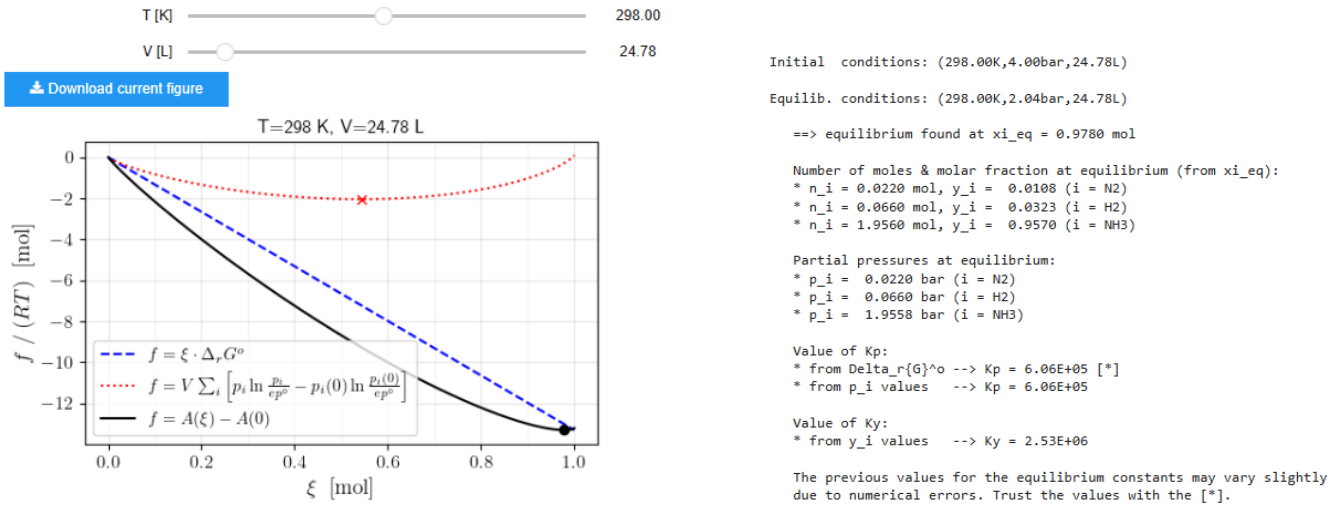


Figure S2. Screenshots showing equilibrium results at constant V and T for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction. Moving the sliders updates the plot interactively.

In the *Statistical-mechanical* section of ***Notebook 2***, the user provides the **PubChem CID** values in order to retrieve initial geometries for the reactants and products. The CIDs used in this example are listed in Table S1. An illustrative output is shown in Fig. S3.

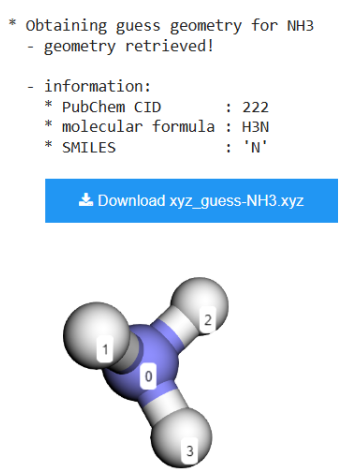


Figure S3. Screenshot showing the retrieved initial structure for NH_3. The view is interactive, allowing the molecule to be inspected from different angles.

For this system, we recommend using the **PBE0** functional and the **6-31G*** basis set. This combination was selected based on benchmarking against experimental data.[1,2] The vibrational frequencies associated with the optimized geometries are listed in Table S2.

Table S2. Comparison of experimental[a] and theoretical vibrational frequencies (cm^{-1}) for H_2, N_2 and NH_3

	H_2	N_2	NH_3
Experimental[a]	4401	2359	950, 1627 (2), 3337, 3444 (2)
PBE0/6-31G*	4451	2498	1131, 1733 (2), 3486, 3627 (2)

[a]Experimental data taken from NIST Chemistry WebBook.[2]

After correcting the symmetry numbers according to Table S1, the thermodynamic quantities reported in Table S3 are obtained.

Table S3. Comparison of experimental[a] and theoretical values of $\Delta_r H°$, $\Delta_r S°$, $\Delta_r G°$, and equilibrium constants for the $N_2 + 3H_2 \rightleftharpoons 2NH_3$ reaction at 298 K

	$\Delta_r H°$ (kJ·mol^{-1})	$\Delta_r S°$ (J·mol^{-1}·K^{-1})	$\Delta_r G°$ (kJ·mol^{-1})
Experimental[a]	-92.22	-198.8	-32.99
PBE0/6-31G*	-90.99	-198.0	-32.00

[a]Experimental data taken from Atkins' *Physical Chemistry* book.[1]

In addition, the temperature dependence of $\Delta_r G°$ shown in Fig. S4 is obtained. Notice that it behaves according to the classical thermodynamics expression, given by Eq. (2) of the main manuscript, if $\Delta_r C°_{p,m}$ is set to -5.71·R.

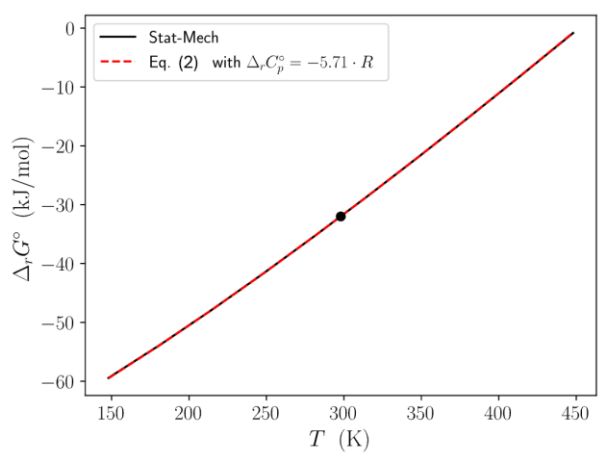


Figure S4. Standard Gibbs energy of reaction for $N_2 + 3H_2 \rightleftharpoons 2NH_3$ as a function of T. Statistical-mechanical results (**black** solid) are compared with the classical thermodynamics expression (**red** dashed) computed using PBE0/6-31G* values of $\Delta_r H°$ and $\Delta_r S°$ at T_{ref} =298 K. A fitted value $\Delta_r C°_p = -5.71 \cdot R$ is used.

3. The $O_2 + 2H_2 \leftrightharpoons 2H_2O$ equilibrium

When running ***Notebook 2***, this reaction is defined by the following input string:

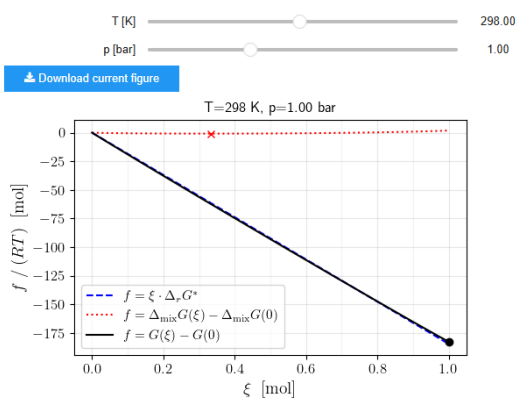
- `O2 + 2 H2 -> 2 H2O`

The remaining input parameters required by the notebook are listed in Table S4.

Table S4. Input data for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction. $\Delta_f H^\circ$ is reported in kJ·mol^{-1}, and S_m° and $C_{p,m}^\circ$ in J·mol^{-1}·K^{-1}. These three values correspond to 298 K.[1]

	$\Delta_f H^\circ$	S_m°	$C_{p,m}^\circ$	PubChem CID	Molecular charge	Number of unpaired electrons	Symmetry number
O_2	0.00	205.138	29.355	977	0	2	2
H_2	0.00	130.684	28.824	783	0	0	2
H_2O	-241.82	188.83	33.58	962	0	0	2

For initial amounts equal to the stoichiometric coefficients of the reactants (*i.e.* 1 mol of O_2 and 2 of H_2, and 0 mol of H_2O), the variation of $G(\xi)$ and of $A(\xi)$ with the extent of reaction, ξ, is shown in Figs. S5 and S6, respectively.




```
Initial  conditions: (298.00K,1.00bar,74.33L)

Equilib. conditions: (298.00K,1.00bar,49.55L)

   ==> equilibrium found at xi_eq = 1.0000 mol

   Number of moles & molar fraction at equilibrium (from xi_eq):
   * n_i = 0.0000 mol, y_i =  0.0000 (i = O2)
   * n_i = 0.0000 mol, y_i =  0.0000 (i = H2)
   * n_i = 2.0000 mol, y_i =  1.0000 (i = H2O)

   Partial pressures at equilibrium:
   * p_i =  0.0000 bar (i = O2)
   * p_i =  0.0000 bar (i = H2)
   * p_i =  1.0000 bar (i = H2O)

   Value of Kp:
   * from Delta_r{G}^o --> Kp = 1.35E+80

   Value of Ky:
   * from Delta_r{G}^* --> Ky = 1.35E+80
```

Figure S5. Screenshots showing equilibrium results at constant p and T for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction. Moving the sliders updates the plot interactively.

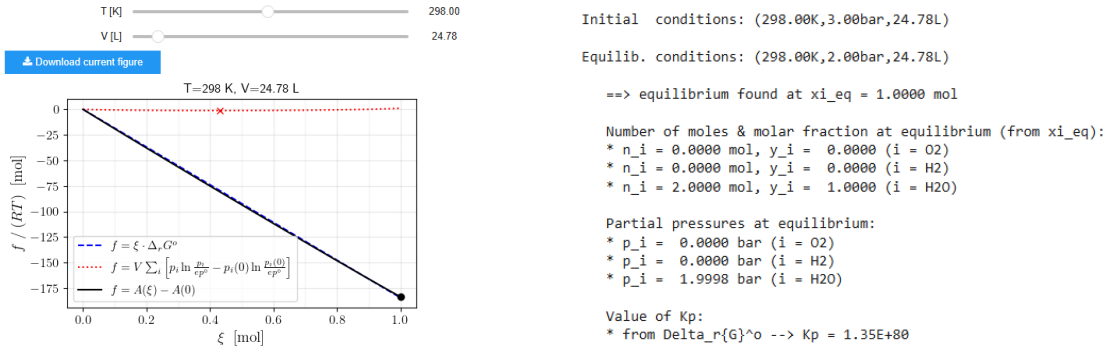


Figure S6. Screenshots showing equilibrium results at constant V and T for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction. Moving the sliders updates the plot interactively.

In the *Statistical-mechanical* section of ***Notebook 2***, the user provides the **PubChem CID** values in order to retrieve initial geometries for the reactants and products. The CIDs used in this example are listed in Table S4. An illustrative output is shown in Fig. S7.

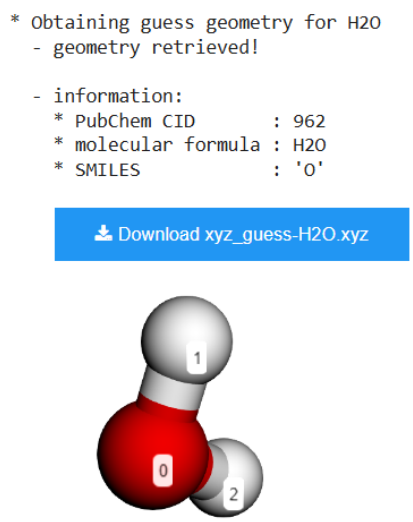


Figure S7. Screenshot showing the retrieved initial structure for H_2O. The view is interactive, allowing the molecule to be inspected from different angles.

For this system, we recommend using the **M06-2X** functional and the **aug-cc-pVDZ** basis set. This combination was selected based on benchmarking against experimental data.[1,2] The vibrational frequencies associated with the optimized geometries are listed in Table S5. Note that electronic-structure calculations for O_2 must be performed with the **correct open-shell configuration** (two unpaired electrons).

Table S5. Comparison of experimental[a] and theoretical vibrational frequencies (cm^{-1}) for H_2, O_2 and H_2O

	H_2	O_2	H_2O
Experimental[a]	4401	1580	1595, 3657, 3756
M06-2X/aug-cc-pVDZ	4477	1771	1614, 3861, 3971

[a]Data taken from NIST Chemistry WebBook.[2]

After correcting the symmetry numbers according to Table S4, the thermodynamic quantities reported in Table S6 are obtained.

Table S6. Comparison of experimental[a] and theoretical values of $\Delta_r H°$, $\Delta_r S°$, $\Delta_r G°$, and equilibrium constants for the $O_2 + 2H_2 \rightleftharpoons 2H_2O$ reaction at 298 K

	$\Delta_r H°$ (kJ·mol^{-1})	$\Delta_r S°$ (J·mol^{-1}·K^{-1})	$\Delta_r G°$ (kJ·mol^{-1})
Experimental[a]	-483.6	-88.64	-457.2
M06-2X/aug-cc-pVDZ	-477.7	-88.58	-451.3

[a]Data taken from Atkins' *Physical Chemistry* book.[1]

In addition, the temperature dependence of $\Delta_r G°$ shown in Fig. S8 is obtained. Notice that it behaves according to the classical thermodynamics expression, given by Eq. (2) of the main manuscript, if $\Delta_r C_{p,m}^{\circ}$ is set to -2.47·R.

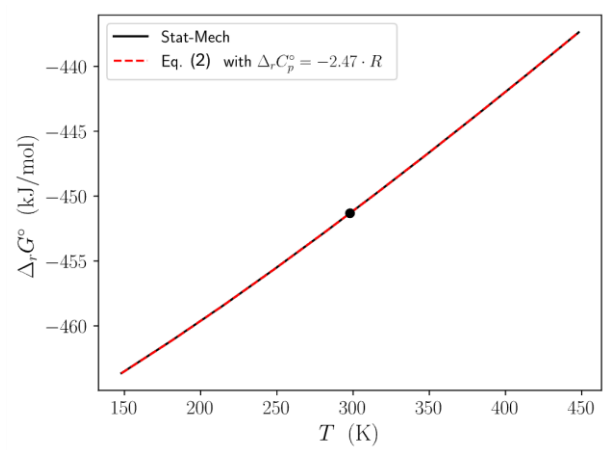


Figure S8. Standard Gibbs energy of reaction for $O_2 + 2H_2 \rightleftharpoons 2H_2O$ as a function of T. Statistical-mechanical results (**black** solid) are compared with the classical thermodynamics expression (**red** dashed) computed using M06-2X/aug-cc-pVDZ values of $\Delta_r H°$ and $\Delta_r S°$ at T_{ref} =298 K. A fitted value $\Delta_r C_p^\circ = -2.47 \cdot R$ is used.

115

4. References

[1] Atkins, P. W.; De Paula, J.; Keeler, J. *Atkins' Physical Chemistry*; Oxford University Press, **2018**.

[2] Linstrom, P. J.; Mallard, W. G. *J. Chem. Eng. Data* **2001**, *46*, 1059-1063.